

**TREE THEORY EXERCISES
DISCRETE MATHEMATICS**



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Exercise:

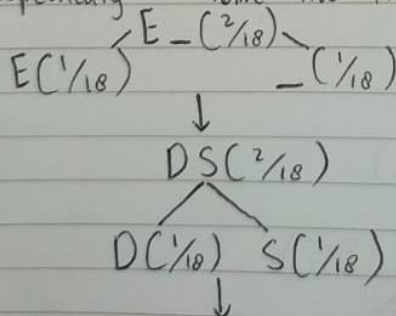
There is a message "MATEMATIKA DISKRIT" in a Script. With the condition that spaces are included in the calculation. Draw the tree, then encode the message using Huffman coding.

Symbols	Frequency	Probability
M	2	$\frac{2}{18}$
A	3	$\frac{3}{18}$
T	3	$\frac{3}{18}$
E	1	$\frac{1}{18}$
I	3	$\frac{3}{18}$
K	2	$\frac{2}{18}$
-	1	$\frac{1}{18}$
D	1	$\frac{1}{18}$
S	1	$\frac{1}{18}$
R	1	$\frac{1}{18}$

* Ascending order

$E(\frac{1}{18})$ $-(\frac{1}{18})$ $D(\frac{1}{18})$ $S(\frac{1}{18})$ $R(\frac{1}{18})$
 $M(\frac{2}{18})$ $K(\frac{2}{18})$ $A(\frac{3}{18})$ $T(\frac{3}{18})$ $I(\frac{3}{18})$

* Repeatedly combine the two smallest frequencies



$RM(\frac{3}{18})$

$RC(\frac{1}{18}) \quad M(\frac{2}{18})$

$\downarrow$
 $KE - (\frac{9}{18})$

$K(\frac{2}{18}) \quad E - (\frac{2}{18})$

$EC(\frac{1}{18}) - (\frac{1}{18})$

$\downarrow$
 $DSRM(\frac{5}{18})$

$DS(\frac{2}{18}) \quad RMC(\frac{3}{18})$

$DC(\frac{1}{18}) \quad SC(\frac{1}{18}) \quad RC(\frac{1}{18}) \quad M(\frac{2}{18})$

$\downarrow$
 $AT(\frac{6}{18})$

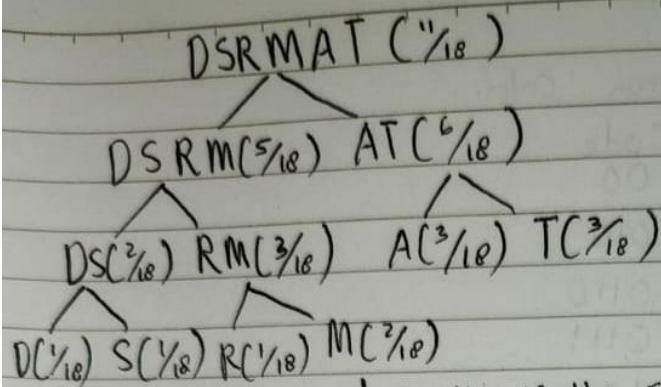
$AC(\frac{3}{18}) \quad T(\frac{3}{18})$

$\downarrow$
 $IKE - (\frac{7}{18})$

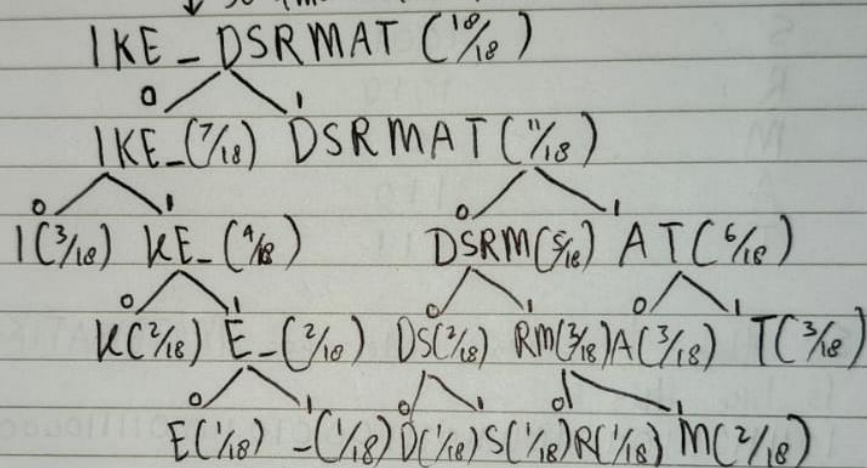
$IC(\frac{3}{18}) \quad KE - (\frac{9}{18})$

$K(\frac{2}{18}) \quad E - (\frac{2}{18})$

$EC(\frac{1}{18}) - (\frac{1}{18})$



↓ So this is the Final tree:



* Construct the Huffman codes

Character	Code
I	00
K	010
E	0110
-	0111
D	1000
S	1001
R	1010
M	1011
A	110
T	111

So the Huffman code for MATEMATIKA-DISKRIT is like this:

101110111011010110111000101100111000001001010
101000111